WEEK 06 • IS321 ENTERPRISE ARCHITECTURE

EA Value and Risk

Making the Case for EA — and Planning for What Can Go Wrong

TODAY'S ROADMAP

What We'll Cover

01	The Value of EA	Three pillars: improved planning, decision-making, and communication
02	Quantifying EA Program Value	Ten measurable ways EA pays for itself
03	Quantifying EA Program Costs	Where the money actually goes, and what changes over time
04	Managing Risk	Four risk types every EA program has to plan for
05	Mitigating Risk	Concrete actions that lower uncertainty
06	AI-Based Risk Analysis	How AI changes risk identification, prediction, and cost optimization

FOUNDATIONS

The Value Side of EA

EA has a unique capability: bringing together views of strategy, business, and technology so the enterprise can see itself in its current and future states — and model how it might respond to unexpected change.



Improved Planning

Enhances both top-down (strategy-first) and bottom-up (program-level) planning, coordinating what would otherwise be disconnected efforts.



Decision Making

An always-current repository gives decision-makers real-time access to coordinated, higher-quality information.



Communications

A standardized language and methodology that especially helps large, geographically dispersed enterprises stay aligned.

PILLAR 1 OF 3

Improved Planning

EA enhances both directions of planning at once — and gives program and project management a baseline of reference documentation for alignment, standards, and configuration.

Top-Down Planning

Begins with strategy and business considerations, enhanced by the holistic, enterprise-wide perspective EA provides.

Bottom-Up Planning

EA coordinates what would otherwise be disparate, separate program-level planning activities across the enterprise.

PILLAR 2 OF 3

Decision Making

EA improves decision-making by providing comprehensive views of current capabilities and plausible future scenarios — revealing exactly what needs to change.

documented consistently with plausible future scenarios in order to pinpoint existing gaps & overlaps that need to be addressed

1 Real-Time Access

An online EA repository, updated at regular intervals, gives decision-makers higher-quality information at whatever level of detail they need.

rapid decisions → reduces delay → allowing meetings to be more effective in the workplace

2 Coordinated Information

Because EA links to other governance areas — capital planning, project management, security — decisions draw on coordinated data, not siloed reports.

where each stakeholder looks at it from different levels of abstraction

isolated

PILLAR 3 OF 3

Communications



"A picture is worth a thousand words." An online EA repository is like a 24×7 gallery of documents and drawings the whole enterprise can draw on.

adapting new innovative tech requires communication channel to articulate it to top leadership, who evaluates it to align it + scale it across enterprise



1

Standardized Language

↳ share common methodology to follow & terminology to communicate with to facilitate flow & their interaction

One methodology and terminology for developing and managing enterprise resources — especially valuable in large, geographically dispersed enterprises with different work cultures.



2

Enhances, Doesn't Stifle, Diversity

how does it cultivate?

EA should augment cultural creativity, not suppress it — by improving alignment of business and technology to strategic goals.

↳ follows a bottom-up approach → uses tech to align strategy



3

Shared Reference in Any Meeting

The same EA materials can be called up in financial planning, R&D, sales reviews, and daily operations — conveying more information in less time.

technology use could cultivate it

Quantifying EA Program Value (1 of 2)

How to translate EA value to the bottom line is one of the biggest questions executives and LOB managers ask. Much of it can be quantified.

Shorter Planning Cycles	A robust online repository of current and future process/resource information speeds up planning.
More Effective Planning Meetings	A common baseline of reference information reduces ambiguity and increases shared understanding.
Shorter Decision Cycles	A logical framework and archiving method drastically cuts the time to crosswalk strategy, business, and technology.
Improved Reference Information	A standardized documentation framework means information is gathered the same way, every time.
Fewer Duplicate Resources	EA visualizes where current resources overlap and where performance gaps exist — the basis for consolidation.

Quantifying EA Program Value (2 of 2)

Reduced Re-Work	A holistic, step-by-step methodology avoids the duplicative or conflicting capabilities that individual programs create.
Better Resource Integration	Comparing current and future business/technology requirements surfaces performance gaps and solutions.
Fewer People in a Process	Business process reengineering eliminates repetitive steps — streamlined processes can mean real payroll savings.
Improved Communication	A common language and central approach reduces misunderstandings of requirements — which itself reduces re-work.
Reduced Cycle Time	Every resource evaluated from the same holistic viewpoint, using the same tools, avoids duplicative capabilities down the line.

THE OTHER SIDE OF THE LEDGER

Quantifying EA Program Costs

Creating an EA for an entire enterprise can be time-consuming, costly, and disruptive. The cost should be viewed across the full program lifecycle: implementation, maintenance, and refreshment.

Staffing & Administration

Chief Architect and EA team salaries; program administration and enterprise tie-ins.

Stakeholder Sessions

Meetings, facilities, and materials for current- and future-state planning sessions.

Repository Technology

Computers, applications, and web development to build and host the EA repository.

Documentation

Interviews and materials to document current views, future views, and the EA Management Plan.

Modeling Tools

Purchase, use, and periodic refreshment of EA modeling applications and computers.

Ongoing Updates

Regular — typically annual — updates to EA documentation and the online repository.

WHAT THE NUMBERS TEND TO SHOW

Characteristics of EA Cost

initial costs exceed maintenance

Initial Cost > Maintenance Cost

establishing initial EA costs more than annual maintenance

Establishing an initial EA costs more than updating and maintaining it — due to the direct and indirect work of building new processes and capabilities, and earning stakeholder support.

trust / buy-in

Where the Money Goes

largest cost: accommodating labor

aligns with staffing

largest cost category

1. **Labor** — trained architects (the largest cost by far)

2. **Supporting technology** — hardware, software, EA tools
   repository & modeling tools

3. **Facility costs** — the EA team's work area and stakeholder meetings
   stakeholder sessions

The full lifecycle cost should be presented to the EA program sponsor so one-time vs. ongoing costs are both clearly understood.

Managing Risk — Four Types

Risk is related to uncertainty — the potential source of a program's failure or underperformance. Managing risk means lowering or eliminating that uncertainty.

Financial

Initial EA setup needs more funding than periodic updates. Maintenance budget cuts can render EA information stale and eventually useless.

Loss of Key Personnel

EA needs architects, analysts, developers, and programmers — losing team members with these skills delays implementation and raises costs.

Lack of Acceptance

EA-level planning replaces older systems-level approaches, which can create tension between program stakeholders and EA stakeholders.

Schedule Delays

Documenting current/future views and building the EA Management Plan is a project with milestones — delays can range from negligible to catastrophic.

Mitigating Risk

Actions that mitigate risk:

- Strengthen executive support for the EA program
- Solidify budgets — don't be the first adopter of new EA tools and techniques
- Ensure trained back-ups exist on the EA team
- Use a detailed EA implementation methodology to guide the whole program
- Apply basic program management skills to address turnover, cost/schedule overruns, performance issues, and stakeholder acceptance

Risk identification and mitigation isn't a one-time activity — it's an ongoing management review item.

NEW SECTION

AI-Based Risk Analysis

Everything above describes how EA programs have managed value and risk for years. AI is now changing how enterprises identify, predict, and price that risk.



Identifying Enterprise Risks



Predictive Risk Modeling



AI-Driven Cost & Value Optimization

AI for Identifying Enterprise Risks



The four risk types on the previous slides are usually surfaced by an architect's manual review. AI can scan the EA repository itself — goals, current/future views, cost records — to flag risk patterns a manual review might miss, such as a Line of Business with no documented back-up architect, or a budget trend that mirrors past programs that stalled.

EXAMPLE

Feeding the EA Management Plan's staffing and schedule sections into a generative AI tool (per PSU's approved ChatGPT/Gemini policy) to ask: "Which Lines of Business have a single point of failure on key personnel, based on this plan?"

AI-BASED RISK ANALYSIS — 2 OF 3

Predictive Risk Modeling

Rather than waiting for a schedule delay or budget cut to happen, predictive models trained on past EA program data can estimate the probability of each risk type materializing — turning "managing risk" from a reactive activity into a forward-looking one.

EXAMPLE

A predictive model trained on prior university IT programs flags that projects entering their second year without an executive sponsor update have historically faced schedule delays 70% of the time — prompting an early check-in for this year's EA program.

AI-Driven Cost and Value Optimization

The ten value-quantification items and six cost categories earlier in this deck are usually estimated by hand. AI can model trade-offs across them simultaneously — for example, weighing the labor cost of an extra architect against the risk-reduction value of having a trained back-up.

EXAMPLE

An AI-assisted scenario model compares two options — hiring a second architect now vs. accepting slower documentation updates — and estimates each option's effect on both program cost and schedule-delay risk.

CHAPTER SUMMARY

Key Takeaways

- EA's value rests on three pillars: improved planning, better decision-making, and clearer communication.
- Ten distinct, largely measurable factors let executives translate EA value into the bottom line.
- EA costs follow a predictable pattern: initial setup costs more than maintenance, and labor is by far the largest cost category.
- Four risk types — financial, personnel, acceptance, and schedule — can be actively managed and mitigated, not just accepted.
- AI-Based Risk Analysis adds three capabilities on top of the traditional approach: identifying risks in repository data, predictive modeling, and cost/value trade-off optimization.

COMING UP NEXT

Further Reading & Next Session

Reading Material

- Bernard, S. (2012). An Introduction to Enterprise Architecture (3rd ed.), Chapter 3 — Primary text
- PSU IS321 Syllabus — AI and EA Tools policy
- Week 4's EA Management Plan (cost & schedule sections referenced above)

Next: Week 7

Models of Organizations and Enterprises — the Leavitt Diamond, the Parsons/Thompson Model, and the Organizational Network Model. Mapped to CLO2.

Questions?